

ECS 409/609

Assignment 7

Conditional Construct

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For code repository of this assignment:  MIPS Assignments

1. Problem 1

Write a MIPS assembly program that takes two number (can be anything oating point or integer) as an input and print maximum between two of them as follows:

32.6 is greater than 25.0

MIPS Assembly Source

```
1 # Assignment 6: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5 # --- Strings for Prompts and Output ---
6 prompt1: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter the first
7             floating-point number: "
8 prompt2: .asciiz "Enter the second floating-point number: "
9 is_greater: .asciiz "is greater than "
10 newline: .asciiz "\n"
11
12 .text
13 .globl main
14 main:
15     li $v0, 4
16     la $a0, prompt1
17     syscall
18
19     li $v0, 6
20     syscall
21
22     mov.s $f1, $f0
23
24     li $v0, 4
25     la $a0, prompt2
26     syscall
27
```

```

28     li $v0, 6
29     syscall
30
31     c.lt.s $f0, $f1
32
33     bc1t num1_is_greater
34
35     li $v0, 2
36     mov.s $f12, $f0
37     syscall
38
39     li $v0, 4
40     la $a0, is_greater
41     syscall
42
43     li $v0, 2
44     mov.s $f12, $f1
45     syscall
46
47     j exit
48
49 num1_is_greater:
50
51     li $v0, 2
52     mov.s $f12, $f1
53     syscall
54
55     li $v0, 4
56     la $a0, is_greater
57     syscall
58
59     li $v0, 2
60     mov.s $f12, $f0
61     syscall
62
63 exit:
64     li $v0, 4
65     la $a0, newline
66     syscall
67
68     li $v0, 10
69     syscall

```

Output Screenshots

Register values and Console

The screenshot displays a MIPS simulator interface. On the left, the 'Int Regs [10]' window shows the following register values:

Register	Value
PC	4194500
EPC	0
Cause	0
BadVAddr	0
Status	805371664
HI	0
LO	0
R0 [r0]	0
R1 [at]	268500992
R2 [v0]	10
R3 [v1]	0
R4 [a0]	268501132
R5 [a1]	2147479220
R6 [a2]	2147479240
R7 [a3]	0
R8 [t0]	0
R9 [t1]	0
R10 [t2]	0
R11 [t3]	0
R12 [t4]	0
R13 [t5]	0
R14 [t6]	0
R15 [t7]	0
R16 [s0]	0
R17 [s1]	0
R18 [s2]	0
R19 [s3]	0
R20 [s4]	0
R21 [s5]	0
R22 [s6]	0
R23 [s7]	0
R24 [t8]	0
R25 [t9]	0
R26 [k0]	0
R27 [k1]	0
R28 [gp]	268468224
R29 [sp]	2147479216
R30 [s8]	0
R31 [ra]	4194328

On the right, the 'Console' window shows the following output:

```
Programmed By: Kunwar Arpit Singh 22185
Enter the first floating-point number: 36.20
Enter the second floating-point number: 67.98
67.98000336 is greater than 36.20000076
```

Below the console window, the assembly code is displayed:

```
[004000a4] 34020002 ori $2, $0, 2 ; 59: li $v0, 2
[004000a8] 46000306 mov.s $f12, $f0 ; 60: mov.s $f12, $f0
[004000ac] 0000000c syscall ; 61: syscall
[004000b0] 34020004 ori $2, $0, 4 ; 64: li $v0, 4
[004000b4] 3c011001 lui $1, 4097 [newline] ; 65: la $a0, newline
[004000b8] 3424008c ori $4, $1, 140 [newline]
[004000bc] 0000000c syscall ; 66: syscall
[004000c0] 3402000a ori $2, $0, 10 ; 68: li $v0, 10
[004000c4] 0000000c syscall ; 69: syscall
```

2. Problem 2

Write an assembly program that takes year as an input from the user and check whether the input year is leap year or not. If it is leap year prompt the message

Input year is a leap year

Otherwise, prompt the message

Input year is not a leap year

MIPS Assembly Source

```
1 # Assignment 6: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5     prompt_year: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter a
        year:"
6     message_is_leap: .asciiz "Input year is a leap year\n"
7     message_not_leap: .asciiz "Input year is not a leap year\n"
8
9 .text
10 .globl main
11
12 main:
13     li $v0, 4
14     la $a0, prompt_year
15     syscall
16
17     li $v0, 5
18     syscall
19
20     move $s0, $v0
21
22     li $t1, 4
23     div $s0, $t1
24     mfhi $t2
25
26     bne $t2, $zero, not_leap
27
28     li $t1, 100
29     div $s0, $t1
30     mfhi $t2
31
32     bne $t2, $zero, is_leap
33
34     li $t1, 400
35     div $s0, $t1
36     mfhi $t2
37
```

```
38     bne $t2, $zero, not_leap
39
40 is_leap:
41     li $v0, 4
42     la $a0, message_is_leap
43     syscall
44     j exit
45
46 not_leap:
47     li $v0, 4
48     la $a0, message_not_leap
49     syscall
50
51 exit:
52     li $v0, 10
53     syscall
```

Output Screenshots

Register values and Console

The image displays two screenshots of a debugger interface, showing register values and console output for a program that checks if a year is a leap year.

Top Screenshot:

- Register Values (Int Regs [10]):**
 - PC = 4194452
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 20
 - LO = 20
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501048
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 0
 - R9 [t1] = 100
 - R10 [t2] = 20
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 2020
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter a year: 2020
 - Input year is a leap year
- Text:**
 - Kernel Text Segment [80000000]..[80010000]
 - [80000180] 0001d821 addu \$27, \$0, \$1 ; 90: move \$k1 \$at # Save \$at

Bottom Screenshot:

- Register Values (Int Regs [10]):**
 - PC = 4194452
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 3
 - LO = 505
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501075
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 0
 - R9 [t1] = 4
 - R10 [t2] = 3
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 2023
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter a year: 2023
 - Input year is not a leap year
- Text:**
 - Kernel Text Segment [80000000]..[80010000]
 - [80000180] 0001d821 addu \$27, \$0, \$1 ; 90: move \$k1 \$at # Save \$at

3. Problem 3

Problem statement

Write an assembly program that determines whether the student is allowed to sit the examination provided his/her attendance is 75%. For the given problem statement, the MIPS assembly program takes the following input: The name of student, Total number of class held and Total class attended by the student. The output format is as follows:

Ajay is allowed to sit in the exam.
or
Ajay is not allowed to sit in the exam.

MIPS Assembly Source

```
1 # Assignment 6: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5     prompt_name: .asciiz "Programmed_By:_Kunwar_Arpit_Singh_22185\n\nEnter_
6         student_name:_"
7     prompt_total: .asciiz "Enter_total_number_of_classes_held:_"
8     prompt_attended: .asciiz "Enter_number_of_classes_attended:_"
9     is_allowed: .asciiz "_is_allowed_to_sit_in_the_exam.\n"
10    not_allowed: .asciiz "_is_not_allowed_to_sit_in_the_exam.\n"
11
12    student_name: .space 256
13
14 .text
15 .globl main
16 main:
17     li $v0, 4
18     la $a0, prompt_name
19     syscall
20
21     li $v0, 8
22     la $a0, student_name
23     li $a1, 256
24     syscall
25
26     la $t0, student_name
27
28 find_newline_loop:
29     lb $t1, 0($t0)
30     beq $t1, 10, replace_newline
31     beq $t1, 0, done_checking
32     addi $t0, $t0, 1
33     j find_newline_loop
34
35 replace_newline:
```

```

36     sb $zero, 0($t0)
37
38 done_checking:
39     li $v0, 4
40     la $a0, prompt_total
41     syscall
42
43     li $v0, 5
44     syscall
45     move $s0, $v0
46
47     li $v0, 4
48     la $a0, prompt_attended
49     syscall
50
51     li $v0, 5
52     syscall
53     move $s1, $v0
54
55     li $t0, 100
56     mult $s1, $t0
57     mflo $t2
58
59     li $t0, 75
60     mult $s0, $t0
61     mflo $t3
62
63     blt $t2, $t3, not_allowed_label
64
65     li $v0, 4
66     la $a0, student_name
67     syscall
68
69     li $v0, 4
70     la $a0, is_allowed
71     syscall
72
73     j exit
74
75 not_allowed_label:
76     li $v0, 4
77     la $a0, student_name
78     syscall
79
80     li $v0, 4
81     la $a0, not_allowed
82     syscall
83
84 exit:
85     li $v0, 10
86     syscall

```


Output Screenshots

Register values and Console

The image displays two screenshots of a MIPS simulator interface, showing register values and console output for two different test cases.

Top Screenshot:

- Register Values:**
 - PC = 4194568
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 3000
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501126
 - R5 [a1] = 256
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 75
 - R9 [t1] = 10
 - R10 [t2] = 3100
 - R11 [t3] = 3000
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 40
 - R17 [s1] = 31
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console Output:**

```
Programmed By: Kunwar Arpit Singh 22185
Enter student name: Arpit
Enter total number of classes held: 40
Enter number of classes attended: 31
Arpit is allowed to sit in the exam.
```
- Assembly Code Snippets:**

```
[00400080] 00028021 addu $16, $0, $2 ; 45: move $s0, $v0
[00400084] 34020004 ori $2, $0, 4 ; 47: li $v0, 4
[00400088] 3c011001 lui $1, 4097 [prompt_attended]; 48: la $a0, prompt_attended
[0040008c] 34240063 ori $4, $1, 99 [prompt_attended]
[00400090] 0000000c syscall ; 49: syscall
[00400094] 34020005 ori $2, $0, 5 ; 51: li $v0, 5
[00400098] 0000000c syscall ; 52: syscall
[0040009c] 00028821 addu $17, $0, $2 ; 53: move $s1, $v0
[004000a0] 34080064 ori $8, $0, 100 ; 55: li $t0, 100
```

Bottom Screenshot:

- Register Values:**
 - PC = 4194568
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 3000
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501159
 - R5 [a1] = 256
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 75
 - R9 [t1] = 10
 - R10 [t2] = 2800
 - R11 [t3] = 3000
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 40
 - R17 [s1] = 28
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console Output:**

```
Programmed By: Kunwar Arpit Singh 22185
Enter student name: Ajay
Enter total number of classes held: 40
Enter number of classes attended: 28
Ajay is not allowed to sit in the exam.
```
- Assembly Code Snippets:**

```
[004000e8] 3c011001 lui $1, 4097 [student_name]; 77: la $a0, student_name
[004000ec] 342400cc ori $4, $1, 204 [student_name]
[004000f0] 0000000c syscall ; 78: syscall
[004000f4] 34020004 ori $2, $0, 4 ; 80: li $v0, 4
[004000f8] 3c011001 lui $1, 4097 [not_allowed]; 81: la $a0, not_allowed
[004000fc] 342400a7 ori $4, $1, 167 [not_allowed]
[00400100] 0000000c syscall ; 82: syscall
[00400104] 3402000a ori $2, $0, 10 ; 85: li $v0, 10
[00400108] 0000000c syscall ; 86: syscall
```

4. Problem 4

Problem statement

Write an MIPS assembly program that takes the marks of the student as an input (in the range of 1-100) and assign the grade. The grading policy are as follows:

Grade: A if marks ≥ 80
Grade: B if $80 < \text{marks} \leq 60$
Grade: C if $60 < \text{marks} \leq 40$
Grade: F otherwise

MIPS Assembly Source

```
1 # Assignment 6: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5 prompt_marks: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter
   marks(0-100):"
6 grade_a: .asciiz "Grade: A\n"
7 grade_b: .asciiz "Grade: B\n"
8 grade_c: .asciiz "Grade: C\n"
9 grade_f: .asciiz "Grade: F\n"
10
11 .text
12 .globl main
13
14 main:
15     li $v0, 4
16     la $a0, prompt_marks
17     syscall
18
19     li $v0, 5
20     syscall
21     move $s0, $v0
22
23     li $t0, 40
24     blt $s0, $t0, grade_F
25
26     li $t0, 60
27     blt $s0, $t0, grade_C
28
29     li $t0, 80
30     blt $s0, $t0, grade_B
31
32 grade_A:
33     li $v0, 4
34     la $a0, grade_a
35     syscall
36     j exit
```

```
37
38 grade_B:
39     li $v0, 4
40     la $a0, grade_b
41     syscall
42     j exit
43
44 grade_C:
45     li $v0, 4
46     la $a0, grade_c
47     syscall
48     j exit
49
50 grade_F:
51     li $v0, 4
52     la $a0, grade_f
53     syscall
54
55 exit:
56     li $v0, 10
57     syscall
```

Output Screenshots

Register values and Console

The image displays two screenshots of a MIPS simulator interface, showing the state of registers and the console output for a program that calculates a grade based on marks.

Top Screenshot:

- Registers:** The **Int Regs [10]** tab is selected. The **PC** register is 4194480. The **R16 [s0]** register contains the value 81.
- Console:** The console window shows the program's output: "Enter marks (0-100): 81" and "Grade: A".
- Code:** The assembly code is visible, showing instructions for loading marks into **\$s0** and calculating the grade based on the value in **\$s0**.

Bottom Screenshot:

- Registers:** The **Int Regs [10]** tab is selected. The **PC** register is 4194480. The **R16 [s0]** register contains the value 79.
- Console:** The console window shows the program's output: "Enter marks (0-100): 79" and "Grade: B".
- Code:** The assembly code is visible, showing instructions for loading marks into **\$s0** and calculating the grade based on the value in **\$s0**.

FP Regs

Int Regs [10]

Data

Text

Int Regs [10]

PC = 4194480
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664

HI = 0
LO = 0

R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501075
R5 [a1] = 2147479220
R6 [a2] = 2147479240
R7 [a3] = 0
R8 [t0] = 60
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 59
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0
R23 [s7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 268468224
R29 [sp] = 2147479216
R30 [s8] = 0
R31 [ra] = 4194328

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 59
Grade: C
|

```

[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall
[004000ac] 3402000a ori $2, $0, 10 ; 56: li $v0, 10
[004000b0] 0000000c syscall ; 57: syscall

```

FP Regs

Int Regs [10]

Data

Text

Int Regs [10]

PC = 4194480
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664

HI = 0
LO = 0

R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501085
R5 [a1] = 2147479220
R6 [a2] = 2147479240
R7 [a3] = 0
R8 [t0] = 40
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 39
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0
R23 [s7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 268468224
R29 [sp] = 2147479216
R30 [s8] = 0
R31 [ra] = 4194328

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 39
Grade: F
|

```

[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall
[004000ac] 3402000a ori $2, $0, 10 ; 56: li $v0, 10
[004000b0] 0000000c syscall ; 57: syscall

```